Ensemble-Based Blood Cancer Stage Classification from Smear Images

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Introduction & Motivation

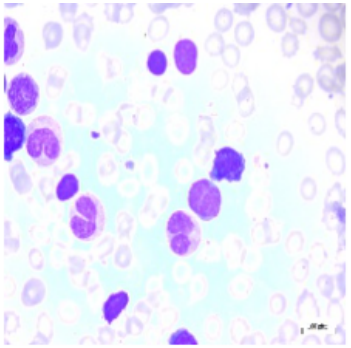
- The global incidence of blood cancer (leukemia) is rising, making rapid and accurate diagnosis essential for improving patient survival.
- Manual examination of peripheral blood smears is time-consuming, labor-intensive, and prone to inter-observer variability.
- Deep learning models offer automation but often function as "black boxes" and require high computational resources (e.g., GPUs), limiting routine clinical use.
- **Proposed Solution:** A transparent, interpretable, and CPU-efficient ensemble learning framework for accurate leukemia stage classification from smear images.

Microscopic Smear Image

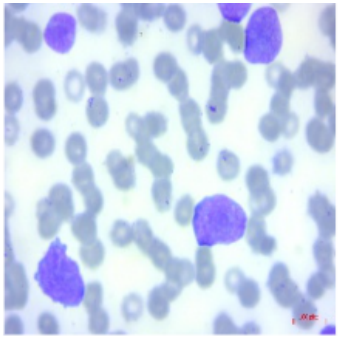
What is a blood smear?
A thin layer of blood spread on a slide, stained, and examined under a microscope to analyze blood cell morphology.

Key components of a white blood cell (WBC):

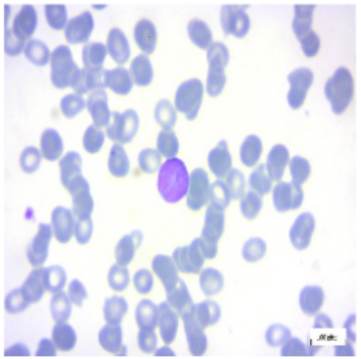
- **Nucleus:** Dark purple region containing genetic material.
- **Cytoplasm** (the lighter area around the nucleus): In healthy cells it is usually pale blue or pink and has a decent amount of space.



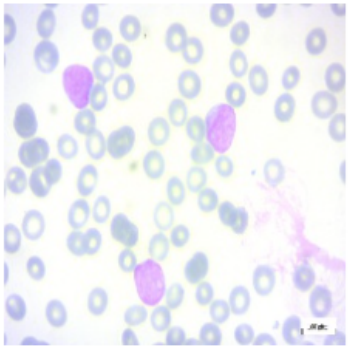
Benign
(Healthy)



Early
(Monitor Closely)



Pre-Cancerous Stage
(Atypical Cells Detected)



Progressive / Malignant Stage
(Blood Cancer Detected)

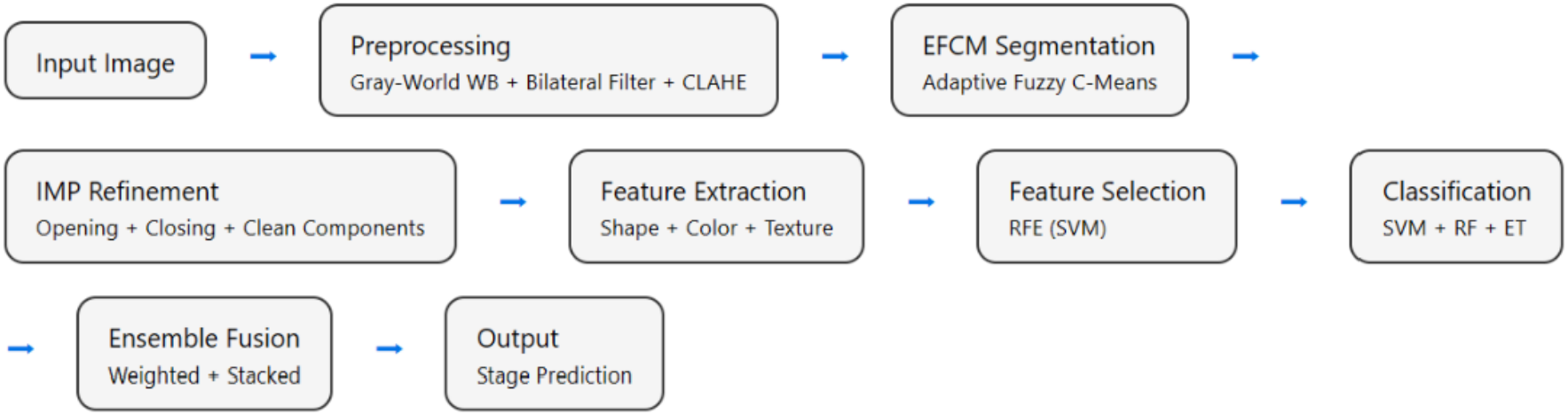
Dataset Description

Source: Mehrad Aria's Leukemia Blood Smear Dataset (Kaggle)

Dataset Overview:

- **Benign (Healthy) — 504 images**
  Round, small nucleus.
  No Cancer Detected — Healthy Sample
- **Early Leukemic — 985 images**
  Slightly enlarged nucleus, reduced cytoplasm.
  Early-Stage Abnormality — Mild Deviations (Monitor Closely)
- **Pre-leukemic — 963 images**
  Large irregular nucleus, atypical morphology.
  Pre-Cancerous Stage — Atypical Cells Detected (Further Evaluation Required)
- **Pro-leukemic / Malignant — 804 images**
  Extremely large nucleus, highly abnormal structure.
  Malignant Stage — Cancer Detected (Immediate Attention Needed)

Pipeline Overview



Stage 1: Preprocessing -Three-Step Image Normalization and Enhancement

1. **Gray-World White Balancing**
   - Eliminates color cast and uneven illumination across RGB channels.
   - Uses the assumption that the average scene color should be neutral gray
   - Formula: $I'_c(x, y) = I_c(x, y) \frac{\mu_{\text{avg}}}{\mu_c}$
2. **Bilateral Filtering**
   - Reduces noise while preserving crucial cellular edges.
   - Combines spatial closeness and intensity similarity to avoid blurring nuclei boundaries.
3. **CLAHE (Contrast Limited Adaptive Histogram Equalization)**
   - Enhances local contrast by applying histogram equalization in small regions (tiles).
   - Prevents over-amplification of noise through a contrast limit.
   - Highlights fine morphological patterns required for feature extraction.

Result: Well-balanced, denoised, enhanced images ready for segmentation.

Stage 2: EFCM Segmentation

Enhanced Fuzzy C-Means (EFCM)

$$u_{ik} = \frac{1}{\sum_{j=1}^{C} \left(\frac{\|x_i - v_k\|^2 + \alpha S_i}{\|x_i - v_j\|^2 + \alpha S_i} \right)^{\frac{1}{m-1}}}$$

Why EFCM is Superior for Smear Images:

- **Spatial Regularization (αS_i):** Uses neighbourhood context to suppress noise and avoid pixel-level misclassification.
- **Stain-Invariant Clustering:** Handles uneven staining or illumination — a common issue in microscopic images.
- **Noise-Resilient Segmentation:** Prevents over-segmentation in low-contrast regions and maintains structural continuity.
- **Cleaner Nucleus–Cytoplasm Differentiation:** Produces smoother, more medically meaningful segmentation masks.

Stage 2: Iterative Morphological Processing

Iterative Morphological Processing (IMP)

- **Opening & Closing Operations:** Remove tiny artifacts, fill small holes, and stabilize region shapes.
- **Boundary Smoothing:** Eliminates jagged edges, producing clinically interpretable contours.
- **Region Cleanup:** Removes false fragments and isolates only the true nucleus and cytoplasm.
- **Consistency Across Samples:** Ensures segmentation uniformity despite variable smear quality.

Final Benefits:

- Clean, well-defined nucleus boundaries.
- Clear separation between nucleus and cytoplasm.
- Preserved morphological integrity for feature extraction.

Stage 3: Feature Extraction (102 Features)

Shape Features

- Area, Perimeter
- Eccentricity
- Circularity: $\frac{4\pi A}{P^2}$
- Solidity, Convexity
- **Hu Moments** (H1–H7)
  - Rotation invariant
  - Scale invariant

Color Features

HSV & LAB spaces:

- Mean (μ)
- Std. deviation (σ)
- Min, Max

$F_c = \{\mu_{h,s,v}, \sigma_{h,s,v}, \min/\max_{l,a,b}$

Captures chromatic variations in nucleus and cytoplasm

Texture Features

- **LBP:** Local Binary Patterns (uniform histograms)
- **GLCM Properties:**
  - Contrast
  - Homogeneity
  - Energy
  - Correlation

Models chromatin texture patterns

Concatenated Feature Vector: $\mathbf{F} = [\mathbf{F}_s \parallel \mathbf{F}_c \parallel \mathbf{F}_t]$ (102 dimensions)

GLCM Texture Formulations

Gray-Level Co-occurrence Matrix (GLCM) Properties:

$$\text{Contrast} = \sum_{i,j} (i-j)^2 P(i,j)$$

$$\text{Energy} = \sum_{i,j} P(i,j)^2$$

$$\text{Homogeneity} = \sum_{i,j} \frac{P(i,j)}{1+|i-j|}$$

$$\text{Correlation} = \sum_{i,j} \frac{(i-\mu_i)(j-\mu_j)P(i,j)}{\sigma_i \sigma_j}$$

Clinical Relevance: These texture descriptors directly correlate with chromatin condensation patterns observed in leukemia progression

Stage 4: Feature Selection (RFE)

Recursive Feature Elimination with Linear SVM

- Eliminate redundant features
- Remove noisy attributes
- Enhance generalization
- Reduce computational cost

Process:

1. Train Linear SVM on all 102 features
2. Rank features by weight magnitude
3. Iteratively remove least important
4. Stop at optimal subset

Result: 48 optimal features retained

Achieves balance between expressiveness and efficiency

$$\mathbf{F}^* = \text{RFE}(\mathbf{F}, n = 48)$$

Stage 5: Ensemble Classification Architecture

Three Base Classifiers:

1. **Support Vector Machine**
   - RBF kernel
   - Nonlinear separation
   - Sharp decision boundaries
2. **Random Forest**
   - Bootstrap aggregation
   - Decision tree ensemble
   - Feature interaction modeling
3. **Extra Trees**
   - Additional randomization
   - Reduced variance
   - Robust predictions

Two Fusion Strategies:

1. Weighted Soft Voting

$$P_{ens}(y = c|x) = \sum_{i=1}^{M} w_{c,i} P_i(y = c|x)$$

Weights ($w_{c,i}$) based on validation F1-scores per class

2. Stacked Ensemble

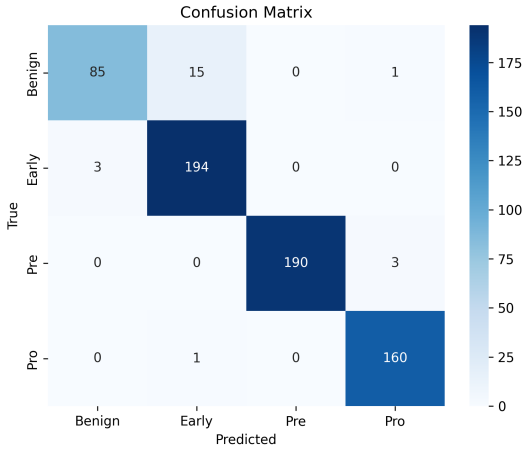
$$\mathbf{z} = [p_{SVM} \parallel p_{RF} \parallel p_{ET}]$$

$$\hat{y} = \arg\max_c \sigma(\mathbf{w}^T \mathbf{z} + b)$$

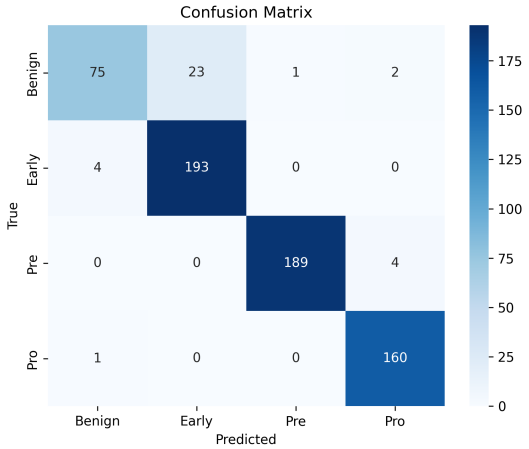
Meta-learner: Logistic Regression Combines out-of-fold predictions

Final Prediction: $\hat{y} = \arg\max_c P_{ens}(y = c|x)$

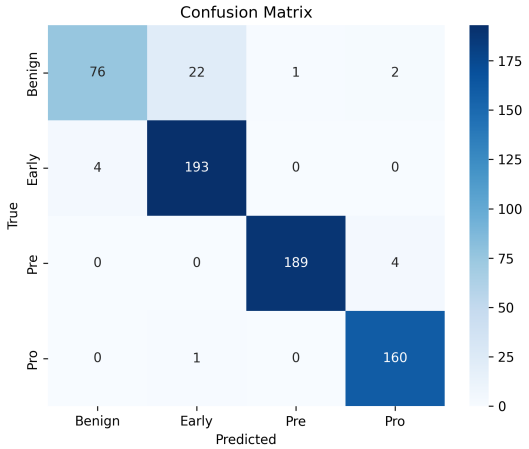
Confusion Matrix Analysis



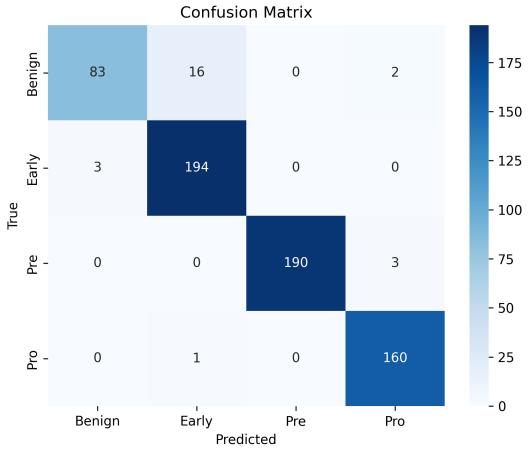
SVM



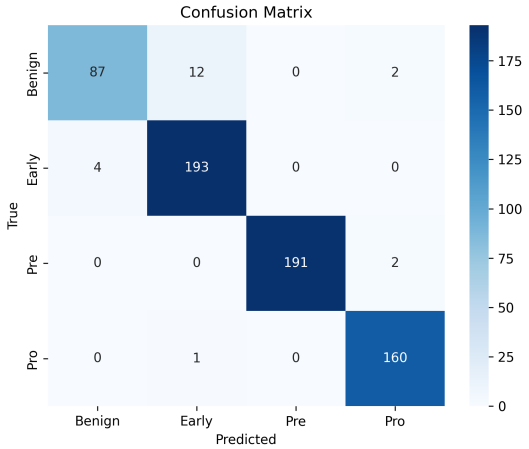
Random Forest



Extra Trees



Weighted Voting



Stacked Ensemble

Classification Performance

Model	Accuracy	Precision	Recall	F1-Score
SVM	0.965	0.966	0.965	0.964
Random Forest	0.946	0.948	0.946	0.945
Extra Trees	0.948	0.950	0.948	0.946
Weighted Voting	0.962	0.963	0.962	0.961
Stacked Ensemble	**0.968**	**0.968**	**0.968**	**0.967**

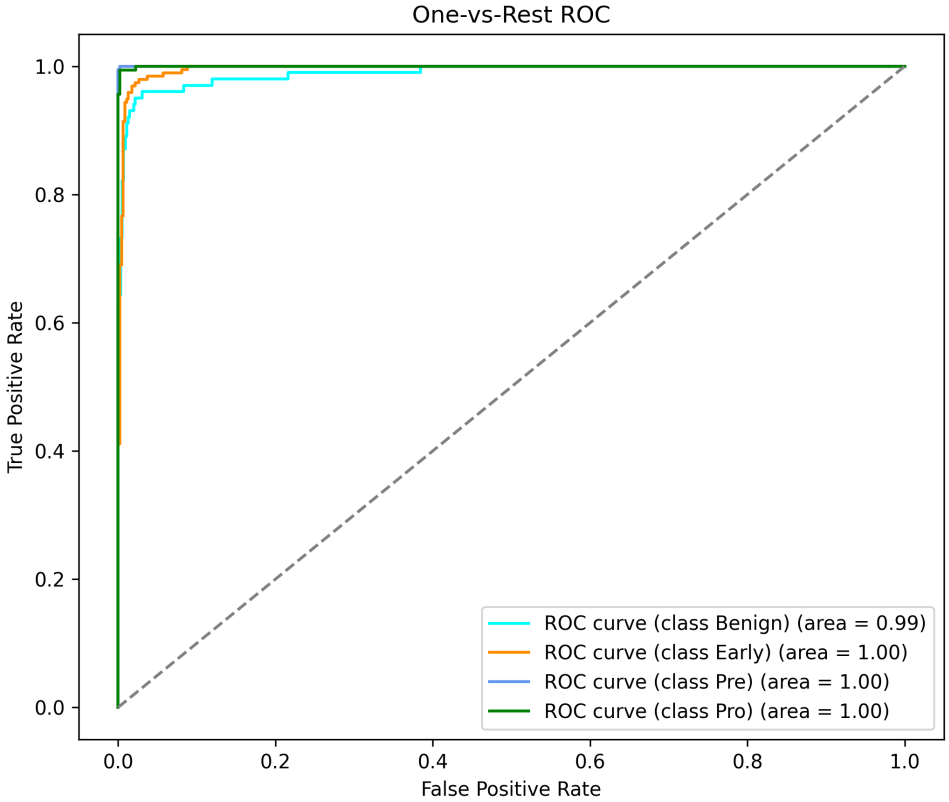
Table: Performance comparison of all classifiers

Key Findings:

- The **Stacked Ensemble** achieves the best overall performance with 96.8% accuracy.
- SVM is the strongest individual model with 96.5% accuracy.
- Ensemble methods consistently outperform standalone classifiers.
- All models maintain balanced precision, recall, and F1-scores.

One-vs-Rest ROC Curve Analysis (Stacked Ensemble)

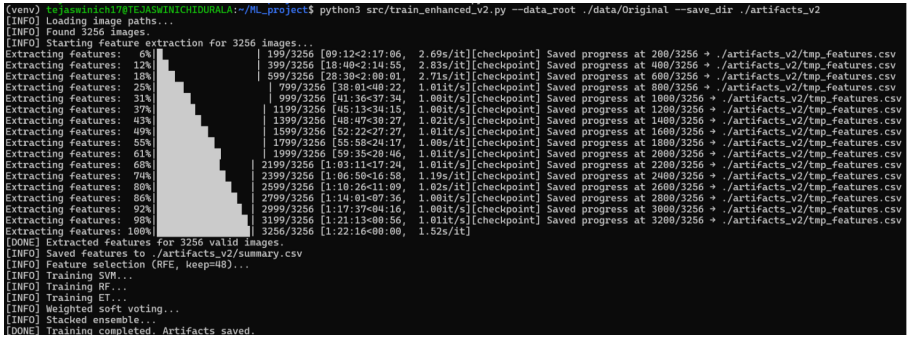
Class	AUC
Benign	0.99
Early	**1.00**
Pre-leukemic	**1.00**
Pro-leukemic	**1.00**
Macro-average	**0.9975**



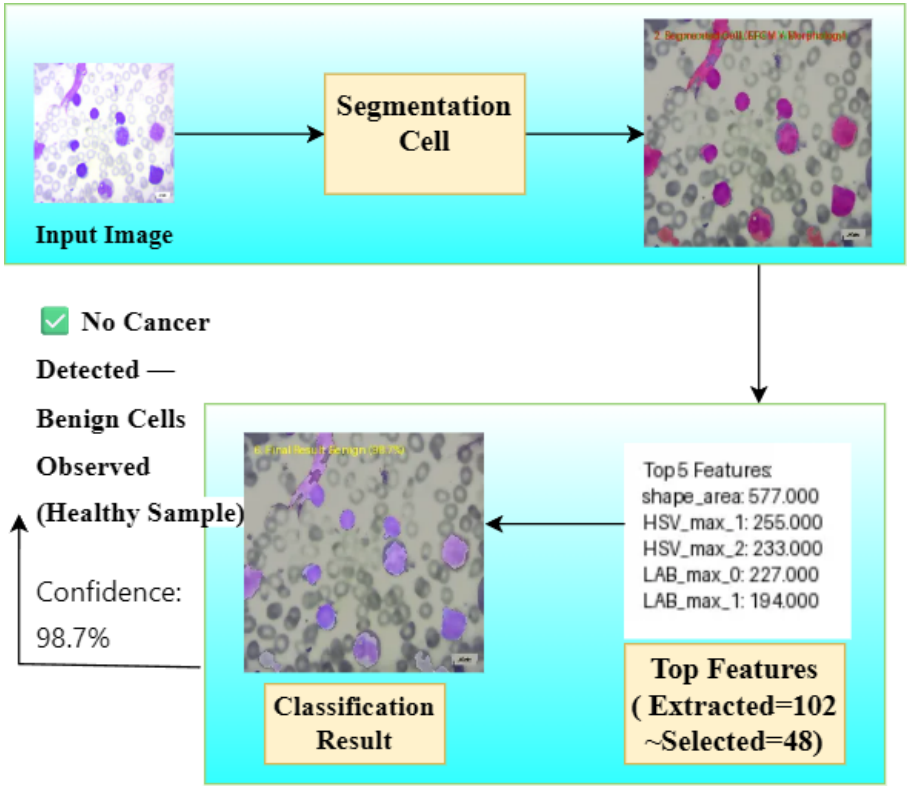
Interpretation:

- Near-perfect class separability across all four stages.
- Three stages achieve an AUC of 1.00.
- Benign (minority class) also achieves an excellent AUC of 0.99.

Screenshots & Demo



Training Interface Dashboard



Gradio-based Prediction UI

Live Demo: HuggingFace Space

Conclusion and Future Work

- The ensemble model improved robustness and generalization.
- Offers potential for real-time screening and early diagnosis support.
- Future work:
  - Transfer learning with larger datasets.
  - Model interpretability (Grad-CAM visualization).
  - Integration into medical decision-support platforms.

References

- S. Rehman et al., "Classification of Acute Lymphoblastic Leukemia using Deep CNNs," *IEEE Access*, 2023.
- R. Jha et al., "Blood Cell Image Classification using Ensemble Learning," *Pattern Recognition Letters*, 2024.
- A. Das et al., "Automated Leukemia Detection using Microscopic Images," *Biomedical Signal Processing and Control*, 2022.

Thank You

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